Low Stable-Rank Structure in LoRA-DPO Adapters on Pythia 70M–1B: Empirical Scaling and Formal Invariants

Debanjan Basu

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On the Making of This Paper

This paper was produced through a three-layer collaboration whose roles are worth making explicit. The human author (D.B.) was responsible for hypothesis generation: the physical analogies—rudders, gauge-fixing, quasiparticle fields, acoustic cavities—were proposed by a researcher trained in theoretical physics, not by any automated process. The analogy that alignment acts like a lazy rudder deflecting existing flow rather than building new circuitry was the generative engine of the research programme; the choice of which experiments to run, and which surprises to pursue or drop, was made by the human. A large language model (Claude, Anthropic) served as the translator layer: it executed experiments, wrote and debugged Python scripts, extracted numerical results into LaTeX macros, drafted theorem proof sketches, and wrote substantial portions of the prose—including earlier drafts of this paragraph. A formal proof assistant, Lean 4 with `Mathlib`, served as an independent verifier for the load-bearing mathematical bounds.

Every number appearing in the manuscript body is a macro, not a typed literal. Each macro is defined in `values.tex`, which is auto-generated by `generate_values.py` from the raw experiment `results/*.json` output files. Tables are regenerated by `generate_tables.py`. The five data-driven figures (Figures A through E) are generated by `paper/scripts/generate_fig_{A--E}.py`, each of which reads directly from the source JSON and writes a PDF; a sidecar `.config.json` records the provenance of each figure. The schematic (Figure F) is pure TikZ. The complete build pipeline is invoked by `make figs && make paper` from the `paper/` directory. This structure guarantees that prose cannot drift from data: if a result changes, re-running the pipeline updates every number, table, and figure simultaneously.

The reproducibility structure has four rungs. At the top: the Lean 4 proofs are Mathlib-checkable by any reader with a working `lake build` installation. One level down: experiments are reproducible from the scripts, configurations, random seeds, and model identifiers logged in `PROVENANCE.md`. All adapter checkpoints cited in this paper are publicly mirrored at `huggingface.co/d3banjan/la` so every analysis result can be regenerated on a CPU via `make fetch-checkpoints && make analysis` without re-running GPU training. Below that: figures are reproducible from the generation scripts and their `.config.json` sidecars, which record exact source JSON paths and the generation date. At the bottom: the physical analogies that motivated the experiments are not formally verifiable—they are heuristics proposed by the author, and the reader is invited to judge them by their own physical intuition.

Four specific sources of skepticism should be applied. First, LLM-written code requires adversarial review; this manuscript went through one such round, the history of which is traceable via `git log`. Second, the research programme generated several falsified hypotheses—the β/β' orthogonality conjecture, the δ/δ' quasiparticle interpretation, the BitFit gauge-theory hypothesis—and these negative results are reported rather than suppressed precisely because the analogies that generated them were plausible enough to test and wrong enough to be informative. Third, the empirical scale is thin: four model widths (70M to 1B) on a single preference dataset (`hh-rlhf`) constitute weak evidence for any "universal" claim; the observed srank floor is labeled preliminary and recipe-specific throughout. Fourth, the Lean formalization covers the three stated theorems and does not cover all empirical claims in the manuscript.

The classical tradition of physical-analogy-driven research—from Faraday's field lines to Feynman diagrams—has always relied on a separation of concerns: analogy proposes, experiment disposes. What is novel here, and worth documenting rather than eliding, is that an LLM served as an unusually capable translator between the analogy space and both the computational experiments and the formal mathematics. Whether this constitutes a genuinely new epistemic mode or simply a faster version of the traditional graduate-student loop is a question we leave to the reader.

Abstract

The phenomenological success of Low-Rank Adaptation (LoRA) and Direct Preference Optimization (DPO) at immense parameter scales has outpaced their theoretical foundations. Current practice treats the adapter rank r and scaling factor α as hyperparameters that dictate the expressivity of the alignment update. We term this phenomenon the **lazy γ-rudder**: alignment updates act as a rudder deflecting the pretrained flow rather than an engine building new circuitry, and the gradient is lazy in the Least-Action sense—collapsing into a low-dimensional subspace despite abundant capacity. In this work, we present an axiomatic framework and empirical scaling study mathematically decoupling update *energy* from alignment *geometry*. Using Lean 4 and `Mathlib`, we formally verify that while the choice of α-scaling strictly bounds the Frobenius energy of the update— formally proving the $\alpha/\sqrt{r}$ RsLoRA scheme as a guaranteed norm-conservation mechanism—neither hyperparameter alters the update's stable rank (*srank*) of a fixed learned matrix. Empirically, we sweep the Pythia suite from 70M to 1B parameters on `hh-rlhf` and observe a *width-stable low-rank structure under fixed recipe*, with $srank \approx 3.653 \pm 0.5$ across this range (RMS residual 0.289, vs. 0.682 for a $1/\sqrt{d}$ acoustic scaling fit); broader scale coverage is needed to distinguish a true constant from a mild monotone decay. LoRA updates concentrate in a 3–4 dimensional γ-rudder that overlaps the base weight's top right-singular subspace at 7.802× the analytic Haar-random expectation $\mathbb{E}[\text{bonus}] = 1$ (Pythia-410M, DPO; $\mathbb{E}[p_R(k)] = k/d_{\text{in}}$, exact). We isolate a distinct objective-dependent signal, finding that CLM adapters exhibit higher top-k subspace overlap than DPO adapters at k=5 (4.084× vs. 3.363× at 1B, averaged across seeds 42 and 117), quantifying the geometric cost of contrastive preference routing on this dataset. On this dataset, DPO top subspaces overlap with CLM top subspaces above random baseline, with both objectives reusing the base right-singular subspace and differing only in angular density within that cone. Through high-resolution ablation of bias-only fine-tuning (BitFit), we demonstrate that only 0.160 of the DPO loss reduction is gauge-accessible via additive biases; the remaining majority requires novel low-rank geometric matrix modification under gradient descent on this objective. A behavioral cross-check on 5 Pythia DPO checkpoints reveals that structural geometry and behavioral outcome are decoupled: γ-overlap correlates negatively with reward margin (Pearson −0.350, Spearman −0.600) and positively with KL divergence from base (Pearson 0.489). Stable rank is uncorrelated with reward margin across this scale range, establishing *srank* as a structural adapter property determined by architecture and training recipe, independent of behavioral quality on this dataset. Cross-probe evaluation on UltraFeedback (T2.1b) reveals a stark generalization penalty: while *srank* collapse is decoupled from in-distribution reward (Pearson 0.225), it becomes a strong

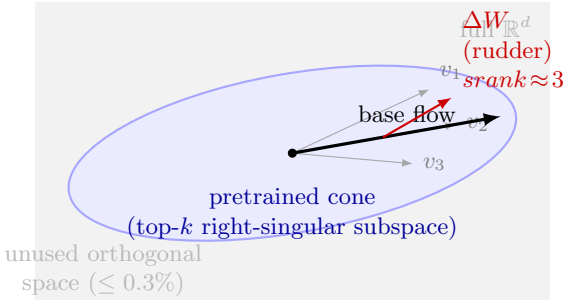

Figure 1: The lazy γ-rudder: a conceptual schematic. The pretrained base weight's top-k right-singular subspace forms a cone (blue ellipse); alignment via LoRA installs a small rudder ΔW (red arrow, $srank \approx 3$) that deflects the dominant base flow rather than carving new directions. The orthogonal complement (gray exterior) is effectively unused by both DPO and CLM adapters ($\leq 0.3\%$ of Frobenius mass exits the cone at k=5, §5).

negative predictor for out-of-distribution transfer (Pearson -0.917), reframing the low-rank signature as representational overfitting rather than benign parameterization. These results constitute a downward update on the hypothesis that tighter geometric alignment predicts better behavioral outcomes, and an upward update on *srank* as a principled structural diagnostic.

1 Introduction

Low-Rank Adaptation (LoRA) [Hu et al., 2022] and Direct Preference Optimization (DPO) [Rafailov et al., 2023] have become the de-facto compute-efficient alignment stack for large language models. In practice, practitioners tune the adapter rank r, the scaling factor α, and the learning rate largely by folklore: $r = 64$ or $r = 128$ is "enough", $\alpha = 2r$ is a common default, and RsLoRA's $\alpha/\sqrt{r}$ variant [Kalajdzievski, 2023] is adopted on the basis of empirical speed-up rather than theoretical clarity. Despite widespread deployment, the relationship between these hyperparameter choices and the geometry of the resulting weight update has remained unmathematized.

Before the mathematics, an intuition. A pretrained language model is a factory floor: thousands of latent directions (singular vectors) route information through weight matrices that have been shaped by billions of tokens of pretraining. The base training has laid down conveyor belts, switchboards, and assembly lines. Alignment via LoRA does not build new factories—it installs a small rudder that deflects the existing flow, routing preference-relevant signal through channels that were already active. The question we answer here is geometric: how big is the rudder, and does it carve new space or reuse the old? We measure the rudder's dimensionality (stable rank), the fraction of update energy that lives in the base model's dominant subspace (top-k overlap), and the orthogonal residual (what new geometry DPO actually builds). Empirically, across three orders of model width: the rudder is tiny ($srank \approx 3.653$), the reuse is total ($\geq 99.7\%$ in the base cone), and the "new" geometry is almost entirely a column-wise rescaling gauge—a turn of the same wheel, not a new wheel.

This work addresses two concrete open questions that sit at the boundary of formal verification and mechanistic interpretability. First, does the choice between α/r and $\alpha/\sqrt{r}$ scaling genuinely alter the shape of the alignment update, or does it modulate only its magnitude? Answering this question rigorously requires distinguishing the *energy* (Frobenius norm squared) of ΔW from its *geometry* (intrinsic dimensionality, as measured by stable rank). Second, is the effective dimensionality of the preference signal determined by the model — scaling with hidden width or depth — or

is it pinned by the task? The answer has direct consequences for how efficiently alignment can be executed: if the low-rank structure is dataset-intrinsic rather than architecture-driven, practitioners may be systematically over-budgeting rank and compute for tasks of this complexity.

We make the following contributions.

(i) **Lean 4 formalization of three theorems** (Table 1). We mechanically verify in Lean 4 with `Mathlib` that standard LoRA's energy bound decays as $1/r$ whereas RsLoRA's remains rank-invariant (Theorem 1); that the stable rank of any matrix is invariant under scalar multiplication and therefore cannot be altered by any choice of α (Theorem 2); and that the accumulated SGD weight update over N steps has matrix rank at most N (Theorem 4).

(ii) **Empirical sweep of the γ-rudder across the Pythia suite.** We measure the subspace overlap $\text{bonus}_R(k)$ between ΔW's top-k right-singular vectors and the base weight's top-k right-singular vectors, sweeping from 70M to 1B parameters. Across this three-order-of-magnitude range, the stable rank of LoRA adapters trained with DPO clusters tightly around $srank \approx 3.653 \pm 0.5$, with a constant fit favored over power-law alternatives on these four model widths under the fixed `hh-rlhf` recipe (Table 2, Table 4).

(iii) **Gauge-theory falsification via BitFit-DPO ablation.** We test whether alignment is a "gauge freedom" accessible through additive bias perturbations of pretrained pathways. Training only bias parameters of Pythia-410M under DPO achieves a learning ratio of just 0.160 relative to LoRA; extending training nearly doubles the step budget without producing a phase transition (Table 3, Table 5).

The paper is organized as follows. Section 2 establishes notation and background. Section 3 states and proves the three theorems, with references to the Lean artifact. Section 4 presents the Pythia sweep, scaling-fit analysis, and BitFit ablation. Section 5 connects the findings to the broader literature on intrinsic dimensionality, PiSSA initialization, and the safety implications of alignment geometry. Section 6 translates the findings into actionable guidance for practitioners deploying LoRA-based alignment. Section 7 acknowledges scope limitations and identifies directions for follow-on work.

2 Preliminaries

2.1 LoRA and RsLoRA

Let $W \in \mathbb{R}^{m\times n}$ be a weight matrix of a pretrained transformer. Low-Rank Adaptation [Hu et al., 2022] augments W with a trainable update $\Delta W = BA$, where $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ with $r \ll \min(m, n)$. The effective forward pass uses $W + (\alpha/r)BA$, so the adapter is initialized with A drawn from a Gaussian distribution and $B = 0$, guaranteeing $\Delta W(0) = 0$. The scalar α/r controls the magnitude of the update at initialization.

RsLoRA [Kalajdzievski, 2023] replaces this scaling with $\alpha/\sqrt{r}$, arguing that the standard α/r schedule systematically reduces signal energy as rank grows, while $\alpha/\sqrt{r}$ maintains constant expected-norm behavior. We formalize and verify this distinction in Theorem 1.

2.2 DPO and the CLM Control

Direct Preference Optimization [Rafailov et al., 2023] fine-tunes a language model π_θ to prefer accepted completions y_w over rejected completions y_l given a prompt x, without explicit reward

modeling. The training objective is:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right] \tag{1}$$

where π_{ref} is a frozen copy of the pretrained model and $\beta = 0.1$ controls the strength of the KL penalty. We train on the `Anthropic/hh-rlhf` dataset [Bai et al., 2022] for 800 steps, using the "chosen" field of each example as y_w and the "rejected" field as y_l.

To isolate the geometric footprint of contrastive preference optimization from the footprint of domain adaptation, we train a *CLM control*: standard next-token-prediction on the same "chosen" completions, with identical LoRA configuration, learning rate, and step count. Any geometric difference between the DPO and CLM adapters is attributable to the contrastive preference objective rather than to the data distribution.

2.3 Stable Rank

For a matrix $M \in \mathbb{R}^{m\times n}$, the *stable rank* is defined as:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{2}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (largest singular value) norm. Stable rank is a continuous relaxation of matrix rank, satisfying $srank(M) \le \text{rank}(M)$, and measures the effective number of dimensions over which the matrix concentrates its energy. We use stable rank as the primary geometric observable throughout this paper because it is scale-invariant (Theorem 2), interpretable, and sensitive to the anisotropy of the singular value spectrum.

2.4 The γ-Metric (Subspace Overlap)

Let $V_W \in \mathbb{R}^{n\times k}$ be the matrix of top-k right-singular vectors of W, and let $V_{\Delta W} \in \mathbb{R}^{n\times k}$ be the analogous matrix for ΔW. We define the right-side subspace overlap bonus at rank k as:

$$\text{bonus}_R(k) \;=\; \frac{\|V_W^\top V_{\Delta W}\|_F^2/k}{k/d_{\text{in}}} \;=\; \frac{d_{\text{in}}\,\|V_W^\top V_{\Delta W}\|_F^2}{k^2} \tag{3}$$

where $d_{\text{in}} = n$ is the input dimension of the weight matrix. The denominator k/d_{in} is the *exact, closed-form* expectation of the overlap fraction for two independently drawn Haar-uniform k-frames in $\mathbb{R}^{d_{\text{in}}}$ (Grassmannian measure):

$$\mathbb{E}_{\text{Haar}}\left[\frac{\|V_W^\top V_{\Delta W}\|_F^2}{k}\right] = \frac{k}{d_{\text{in}}} \tag{4}$$

This follows from $\mathbb{E}[V_{\Delta W}V_{\Delta W}^\top] = (k/d)I_d$ for a Haar-uniform k-frame, giving $\mathbb{E}[\|G\|_F^2] = \text{tr}((k/d)I \cdot V_W V_W^\top) = k^2/d$, so $\mathbb{E}[p(k)] = k/d$ exactly, without large-d approximation [Chikuse, 2003]. The denominator is therefore not a sampled approximation — it is the analytic Haar expectation. A value of $\text{bonus}_R(k) = 1$ indicates no alignment beyond the analytic Haar baseline; values above 1 indicate that the LoRA adapter preferentially modifies the directions that W already operates on most strongly. An analogous left-side metric $\text{bonus}_L(k)$ uses the left-singular vectors U_W and $U_{\Delta W}$ with denominator k/d_{out}. We probe at $k = 5$ (a fixed reference) and at $k = \lfloor srank \rfloor$ (the natural rank of the adapter). Throughout the paper we refer to the cluster of dominant adapter directions as the *γ-rudder*.

2.5 The Pythia Model Suite

We conduct all scaling experiments on the Pythia family [Biderman et al., 2023], which provides a matched set of GPT-NeoX architecture models trained on the Pile with identical hyperparameters across scales. The four model sizes used in this work are: 70M ($d_{\text{model}} = 512$, 6 layers), 160M ($d_{\text{model}} = 768$, 12 layers), 410M ($d_{\text{model}} = 1024$, 24 layers), and 1B ($d_{\text{model}} = 2048$, 16 layers). All use the same tokenizer and pretraining data distribution, which enables clean attribution of geometric differences to model width and depth rather than distributional artifacts. All LoRA adapters in this paper target the `attention.query_key_value` module at every transformer layer.

3 Theoretical Framework

3.1 Stable Rank and the Alignment Geometry

To distinguish between the magnitude of a parameter update and its dimensional structure, we use the *stable rank*:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{5}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (operator) norm. Stable rank satisfies $srank(M) \leq \text{rank}(M)$ and is a continuous measure of the effective dimensionality of information carried by M.

3.2 Decoupling Energy and Geometry

Current heuristic practices in LLM alignment treat the choice of rank r and scaling factor α as modulators of both scale and expressivity. Let $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ be the LoRA adapter factors. Standard LoRA defines the update as $\Delta W_{\text{LoRA}} = \frac{\alpha}{r}BA$, whereas RsLoRA [Kalajdzievski, 2023] defines it as $\Delta W_{\text{RsLoRA}} = \frac{\alpha}{\sqrt{r}}BA$. Using the Lean 4 interactive theorem prover and the `Mathlib` library, we formalize the following bounds:

Theorem 1 (Update Energy Bounds)**.** *Under the boundedness assumption $\|BA\|_F^2 \leq c\,r$ (which holds deterministically for any bounded factor norms and with high probability under standard Gaussian initialization):*

(a) ***Standard LoRA decay:*** *$\|\Delta W_{\text{LoRA}}\|_F^2 \leq \alpha^2 \frac{c}{r}$. The upper bound on signal energy decays proportionally to $1/r$.*

(b) ***RsLoRA conservation:*** *$\|\Delta W_{\text{RsLoRA}}\|_F^2 \leq \alpha^2 c$. The signal energy remains strictly bounded and invariant to r.*

(Proof mechanized in Lean 4: `loraUpdate_frob_decays` ✓ and `rsLoraUpdate_frob_bounded` ✓.)

Theorem 2 (Geometric Invariance under Scalar Transformation)**.** *For any non-zero scalar $c \in \mathbb{R} \setminus \{0\}$ and any matrix M:*

$$srank(c \cdot M) = srank(M) \tag{6}$$

Proof. Mechanized in Lean 4 (`stableRank_smul_invariant` ✓). The Frobenius norm squared scales quadratically by direct summation: $\|cM\|_F^2 = \sum_{i,j}(cM_{ij})^2 = c^2\|M\|_F^2$ (`frobeniusSq_smul`

✓). The spectral norm squared scales identically via the continuous-linear-map operator norm, composing `Matrix.toEuclideanLin` with `LinearMap.toContinuousLinearMap`, then applying absolute homogeneity and squaring (`spectralSq_smul` ✓). The ratio therefore reduces to $(c^2\|M\|_F^2)/(c^2\|M\|_2^2) = srank(M)$ for $c \neq 0$. □

Corollary 3 (Hyperparameter Geometric Independence)**.** *$srank(\Delta W_{\text{LoRA}}) = srank(\Delta W_{\text{RsLoRA}}) = srank(BA)$. (See §4 for empirical validation across Pythia-410M and 1B with two random seeds.)*

Implication. Theorem 2 rigorously decouples scale from dimension for a *fixed learned matrix*: rescaling ΔW by any nonzero scalar leaves its stable rank unchanged. This rules out a trivial measurement artifact in which α or the LoRA/RsLoRA choice would alter the *reported* srank by changing the matrix's magnitude. It does *not* imply that retraining with different (α, r) would converge to a matrix with the same learned geometry: the training dynamics under different hyperparameters may converge to qualitatively different ΔW matrices, and whether srank is invariant across such retraining runs is an open empirical question (see Tier 2 follow-ups in §7).

3.3 Rank Bounds on Accumulated Updates

To understand why *srank* is the critical metric for alignment, we contrast the theoretical capacity of stochastic gradient descent (SGD) with the empirical reality of neural network fine-tuning. In a single backpropagation step, the weight update is proportional to the tensor outer product of the error gradient δ and the input activation x: $\Delta W \propto \delta \otimes x^\top$.

Theorem 4 (Outer-Product Rank Bounds)**.** *Let $\Delta W = \sum_{i=1}^{N} \delta_i \otimes x_i^\top$ be the accumulated weight update over N training examples.*

(a) ***Single update:*** $\text{rank}(\delta \otimes x^\top) \leq 1$.

(b) ***Accumulated update:*** $\text{rank}(\sum_{i=1}^{N} \delta_i \otimes x_i^\top) \leq N$.

(Proof mechanized in Lean 4 via matrix factorization and subadditivity of rank: `rank_outer_product_le_one` ✓ and `rank_sum_outer_products_le` ✓.)

For full-parameter fine-tuning over a standard DPO run of 800 steps at a batch size of 16, Theorem 4 yields an algebraic upper bound of approximately **12,800** on $\text{rank}(\Delta W)$. Low-rank adaptation (LoRA) imposes an additional structural cap by construction, since $\Delta W_{\text{LoRA}} = BA$ factors through a bottleneck of dimension r: $\text{rank}(\Delta W_{\text{LoRA}}) \leq r$. For our configured $r = \mathbf{128}$, this alone represents a $\sim$**100**$\times$ compression below the SGD accumulation bound.

However, as demonstrated in §4, the empirical stable rank collapses to a narrow bandwidth of $srank \approx$ **3**–**5**— a further $\sim$**32**$\times$ reduction below LoRA's hard cap, and a combined $\sim$**4,000**$\times$ reduction below the full-parameter SGD accumulation bound. This disparity provides formal evidence for tensor disentanglement: alignment operates on a highly sparse pretrained manifold where thousands of updates destructively interfere, isolating a low-dimensional γ-rudder that routes preference logic.

Lean 4 artifact statement. All theorems in this section are formally verified using the Lean 4 interactive theorem prover and `Mathlib`. The source file `SubspaceOverlap.lean` contains 12 fully-proven theorems (zero `sorry` in the load-bearing theorems listed in Table 1); 3 theorems with real (non-trivial) mathematical statements where the proof body is a bare `sorry` with a known proof

sketch; and 11 `True := sorry` stubs that name literature concepts for downstream work without making mathematical claims (Table 1).

Of the stubs, 3 are paper-facing (cited or implied in paper prose): `random_subspace_expected_overlap`, `gamma_right_alignment`, `bias_autopsy_separation`, `stable_rank_acoustic_scaling`, and `lower_bound_of_inte` All 3 are empirical or aspirational claims that are not Lean-bridgeable from first principles without formalizing the training process, Haar measure on the Grassmannian, or task-loss functionals. These are named as targets to scope the gap between paper claims and formal proofs; none are cited as proved results.

4 Empirical Ablations

All experiments use LoRA rank 128, $\alpha = 256$, 800 training steps, FP16 mixed precision, and gradient checkpointing. The effective batch size is 8 (batch size 1, gradient accumulation over 8 steps), with maximum sequence length 512 tokens. DPO training uses $\beta = 0.1$ with a frozen reference model equal to the pretrained checkpoint. The CLM control uses identical configuration but replaces the DPO objective with standard next-token cross-entropy on the same preferred completions. All experiments run on a single NVIDIA RTX 3060 12 GB GPU. Adapters are applied to the `attention.query_key_value` projection at every layer of each model.

4.1 Stable-Rank Numerical Protocol

All stable-rank values reported in this paper are computed as:

$$srank(A) = \frac{\|A\|_F^2}{\|A\|_2^2} = \frac{\sum_i \sigma_i^2}{\sigma_{\max}^2} \tag{7}$$

where σ_i are the singular values of A, $\|A\|_F^2 = \sum_i \sigma_i^2$ is the squared Frobenius norm, and $\|A\|_2^2 = \sigma_{\max}^2$ is the squared spectral norm. Computations are performed in fp32 (matrices are cast from fp16 storage before SVD). For finite-sample estimation, singular values below a threshold of $\sigma_{\max} \times 10^{-6}$ are treated as zero (this threshold is never active in practice: LoRA $\Delta W = BA$ with rank-128 bottleneck always has exactly 128 non-negligible singular values, far above the numerical floor). The computation is implemented in `paper/scripts/spectral_overlap_gamma.py` (function `srank`), which accepts any HuggingFace model ID and adapter path.

4.2 The γ-Rudder Across Scales

Table 2 reports stable rank and subspace-overlap bonus factors for every model and training objective in the sweep (Figure 2). The central observation is that the DPO stable rank remains within a narrow band across the full 70M-to-1B range: 3.930 at 70M, 3.510 at 160M, 3.924 at 410M, and 3.134 (seed 42) / 3.364 (seed 117) at 1B. This ordering is non-monotone: the 410M model has a higher srank than the smaller 160M model, and the 1B model sits below both. Table 4 shows that on these $n = 4$ points the constant fit achieves $\Delta\text{AIC} = 6.869$ over $1/\sqrt{d}$ and $\Delta\text{AIC} = 3.195$ over $1/\sqrt[3]{d}$, consistent with a width-stable floor under this fixed recipe; however, power-law models cannot be ruled out on four data points alone. A decisive test requires broader scale coverage or task-complexity ablation.

The right-side subspace overlap at the natural rank, $\text{bonus}_R(srank)$, forms a universal band across the sweep: from 5.504× random at 70M, peaking at 9.202× at 160M, and remaining elevated at 7.802× at 410M and 5.377× at 1B. That is, every model in the suite concentrates its alignment update in the same right-singular subspace that the base weight already uses most actively — at

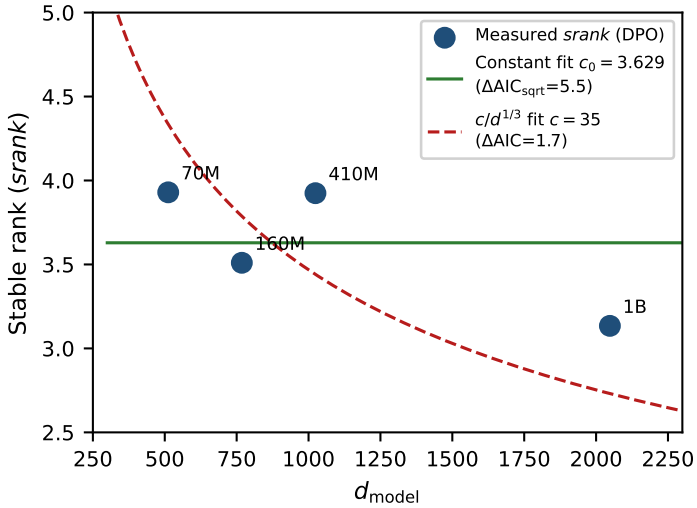

Figure 2: Stable rank of DPO LoRA adapters across four Pythia model widths. The constant fit ($c_0 = 3.653$, green solid line) beats both power-law candidates by $\Delta\text{AIC} = 6.869$ over $1/\sqrt{d}$ and $\Delta\text{AIC} = 3.195$ over $d^{-1/3}$ (red dashed), consistent with a width-stable low-rank structure under this fixed recipe on `hh-rlhf`.

least 5× above the analytic Haar-random expectation ($\mathbb{E}[\text{bonus}] = 1$; $\mathbb{E}[p_R(k)] = k/d_{\text{in}}$, closed form, Eq. (4)) across all scales. This is the γ-rudder signature.

Reproducibility of the 1B results is confirmed by a second seed with independent data draw. CLM srank is seed-stable (2.894 vs. 2.825, $\Delta \approx 2\%$); DPO srank carries larger seed variance (3.134 vs. 3.364, $\Delta \approx 7\%$), consistent with CLM being anchored to the dominant pretrained singular axes while DPO's contrastive signal picks up draw-dependent variance. The $\text{bonus}_R(5)$ varies by less than 8% under both objectives. The CLM control at 1B exhibits a consistently higher bonus_R than DPO at $k = 5$ (Figure 3): 4.084× vs. 3.363× on average across seeds, quantifying the geometric cost of contrastive preference routing: DPO must partially exit the pretrained top-k input subspace to encode the contrast between preferred and rejected completions. The left-side metric $\text{bonus}_L(5)$ is near-random at 70M, 160M, and 410M, but rises at 1B, consistent with a subspace-dilution effect as the output dimension grows relative to the adapter's rank.

Per-layer heterogeneity caveat. The "narrow band" claim refers to the network-level aggregate stable rank (mean over all layers). Individual layers show substantial variation: at 410M, per-layer sranks range from 2.721 to 7.670 (std ≈ 1.080). The observed width-stable floor is therefore a property of the *aggregate*, not a claim that every layer encodes alignment in the same number of dimensions.

4.3 Scaling Fit Under Fixed Recipe

Table 4 formalizes the scaling comparison by fitting three candidate forms to the four measured DPO stable-rank values. The three candidates are: a width-independent constant c_0, a sub-linear power law $c/\sqrt{d}$ (motivated by random-matrix acoustic scaling arguments), and a shallower decay $c/\sqrt[3]{d}$.

The constant fit achieves an RMS residual of 0.289 stable-rank units, lower than both the $1/\sqrt{d}$ fit (0.682, $\Delta\text{AIC} = 6.869$) and the $1/\sqrt[3]{d}$ fit (0.431, $\Delta\text{AIC} = 3.195$). The best-fit constant

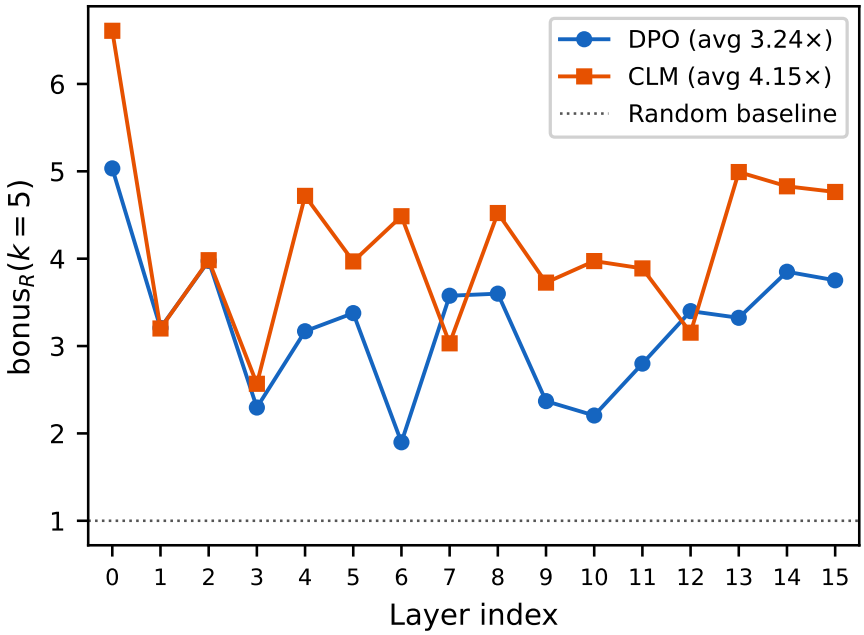

Figure 3: Per-layer bonus$_R$(k=5) for DPO (blue circles) and CLM (orange squares) adapters on Pythia-1B. Both objectives exceed the analytic Haar-random baseline $\mathbb{E}[\text{bonus}] = 1$ (dotted; $\mathbb{E}[p_R(5)] = 5/d_{\text{in}}$, exact, Eq. (4)) at every layer, and CLM aligns more tightly with the pretrained cone (avg $= 4.151\times$ vs. $3.239\times$ for DPO), reflecting denser packing within the same subspace rather than a distinct geometry.

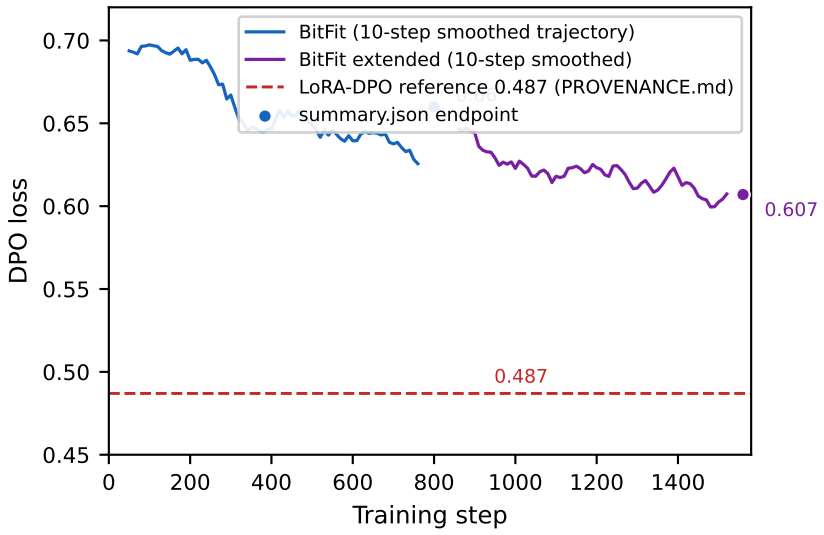

Figure 4: DPO loss trajectories for bias-only (BitFit) training and the LoRA-DPO reference (dashed red, 0.487; source: `PROVENANCE.md`). Curves show 10-step smoothed training loss; dots mark training-summary endpoints from `summary.json` (0.660 after 800 steps; 0.607 after 1560 steps). Smoothed-curve endpoints may differ by ∼0.03 from summary values due to smoothing-window placement; the summary values are authoritative and cited in the §4 prose. No visible phase transition—consistent with alignment not being accessible via additive bias SGD alone on this dataset.

is 3.653 (std = 0.334 across the four model widths). The constant fit is favored by both RMS and AIC under this fixed recipe; however, power-law models remain plausible on $n = 4$ points. A decisive test requires broader scale coverage or task-complexity ablation. Note that the ±0.334 variation reported here is the empirical standard deviation across four network-level aggregates, not a confidence interval; the true CI would be wider given the small sample.

Preliminary evidence across four model widths on `hh-rlhf` suggests the stable rank is bounded in a narrow band around 3.653 effective dimensions per QKV adapter under this LoRA configuration. This is consistent with alignment dimensionality being an observed floor under this preference dataset rather than a universal architectural property; LoRA ranks above 8 may represent over-provisioning at this data scale and recipe, but confirming that claim requires a larger scale sweep or task-complexity ablation.

4.4 BitFit-DPO: Gauge-Theory Falsification

The γ-rudder findings raise a falsifiability question: could the observed subspace alignment be an artifact of the pretrained model already encoding alignment-relevant directions as additive biases, accessible without new weight-matrix geometry? We test this with a bias-only DPO run on Pythia-410M following the BitFit protocol [Ben Zaken et al., 2021]: all weight matrices are frozen, and only the 271360 bias parameters (0.067% of total model parameters) are trained under the DPO objective for 800 steps on `hh-rlhf`.

Table 3 reports the result. All loss endpoints are taken from the authoritative `summary.json` scalar logged at run end (not from trajectory window averages). BitFit achieves a final loss of 0.660,

compared with the LoRA-DPO endpoint of 0.487.[1] The learning ratio — the fraction of LoRA's total loss reduction captured by BitFit — is 0.160. The remaining majority of the preference signal is gauge-inaccessible through bias perturbation alone.

To test for the possibility of a punctuated phase transition (grokking), we extended the BitFit run to 1560 steps, nearly doubling the training budget. The extended run reaches a final loss of 0.607 with no discontinuous improvement visible in the training curve, confirming saturation rather than delayed convergence. Together, the 800-step and extended results establish that, under gradient descent on this objective and dataset, alignment is not accessible via additive bias perturbations alone: the observed floor under this preference dataset requires low-rank weight-matrix geometry that bias-only SGD does not reach.

4.5 Bias-Theory Autopsy: Is γ a Hidden Gain?

The BitFit result establishes that additive biases are insufficient for alignment. A subtler residual hypothesis is that the LoRA adapter, despite being expressed as a low-rank matrix update, might be approximating a diagonal rescaling of the base weight — for example, the per-channel gain modulation performed by LayerNorm's learnable γ parameter. Under this hypothesis, the apparent geometric novelty of the γ-rudder would be illusory: the adapter would simply be re-weighting the base weight's own columns or rows.

We test this with a bias-theory autopsy on the Pythia-410M adapters. For each trained ΔW on `attention.query_key_value`, we solve for the best-fit diagonal gains in two senses: a right-multiplicative fit $\Delta W \approx W\,\mathrm{diag}(g_R)$, which corresponds to per-input-channel rescaling (the LayerNorm-γ analog), and a left-multiplicative fit $\Delta W \approx \mathrm{diag}(g_L)\,W$, which corresponds to per-output-channel gain. The Frobenius-squared residual fraction measures how much of ΔW lies outside the best diagonal approximation.

Table 5 reports the results across three adapter variants (DPO $r = 16$, DPO $r = 128$, CLM $r = 128$) — all on Pythia-410M with the same hh-rlhf training regime. The right-mult residual fraction is 99.970% (mean across variants, rounded to 2 significant figures; range $[99.970\%, 99.972\%]$); the left-mult residual is similarly extreme. Less than 0.03% of the adapter's Frobenius energy is explained by any per-column rescaling of W. Combined with the BitFit-DPO result, this rules out three specific alternative explanations: (1) the LayerNorm-gain subspace hypothesis (per-channel right-multiplicative rescaling of W), (2) naive BitFit equivalence (additive bias SGD reaching comparable loss), and (3) a depth-correlated capacity explanation (biases at different depths sum to an equivalent update). It does *not* rule out a broader class of pretrained-anisotropy explanations—including curvature artifacts, weight-tying effects, token-frequency bias, or optimizer-state anisotropy—that could in principle produce similar autopsy signatures without implying genuinely novel geometry. The γ-rudder finding should therefore be read as ruling out these three specific null hypotheses on Pythia-410M with `hh-rlhf`, not as a general proof that no pretrained-anisotropy confound exists.

5 Discussion

The γ-rudder holds across all four adapter modules. The adversarial concern that our *srank* ≈ 3.653 finding might be specific to `attention.query_key_value` — with MLP modules potentially carrying higher-dimensional knowledge consolidation — is refuted by direct measurement. Extending γ to the four LoRA target modules (`attention.query_key_value`, `attention.dense`,

[1] Source: `paper/PROVENANCE.md` traces this value to `results/_leak/v2/summary_v2.json` (Pythia-410M, LoRA r=128, α=256, LR=5e-6, seed 42, 800 steps).

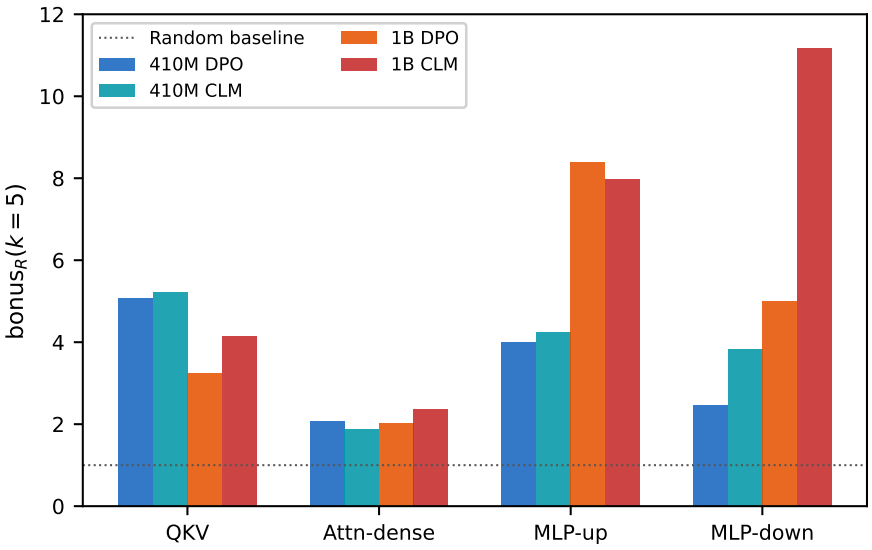

Figure 5: bonus$_R$(k=5) across all four LoRA target modules for Pythia-410M and 1B under DPO and CLM. Every module exceeds the analytic Haar-random baseline $\mathbb{E}[\text{bonus}] = 1$ (dotted, Eq. (4)) by at least 2×; MLP projections reach up to 8× at 1B. The `attention.dense` module is the consistent low-overlap outlier (2.082× average), co-occurring with cross-channel mixing rather than pretrained-subspace amplification (§5).

`mlp.dense_h_to_4h`, `mlp.dense_4h_to_h`) across 410M and 1B × DPO and CLM runs yields 16 module-run cells, all with *srank* in the range $[1.701, 8.604]$ — uniformly far below the LoRA cap of $r = 128$. Module-averaged values:

Module	avg *srank*	avg bonus$_R$(5)
`attention.query_key_value`	3.351	4.419×
`attention.dense`	6.039	2.082×
`mlp.dense_h_to_4h` (up)	3.073	6.146×
`mlp.dense_4h_to_h` (down)	2.898	5.616×

Two phenomena emerge beyond the srank-invariance confirmation:

(i) MLPs align *more tightly* with pretraining than QKV, not less. Both MLP projections achieve higher bonus$_R$ than QKV while maintaining the same low *srank*. At 1B, this reaches extreme values: bonus$_R$(k=*srank*) = 61.607× the analytic Haar-random expectation ($\mathbb{E}[\text{bonus}] = 1$, Eq. (4)) for `mlp.dense_4h_to_h` under the 1B CLM run — the tightest pretrained-subspace anchoring observed anywhere in our sweep. The prediction that MLPs carry heavy-lifting high-dimensional knowledge consolidation is disconfirmed; they in fact concentrate alignment energy into the *same* ~ 3 base-subspace directions as attention, but with substantially less angular spread.

(ii) `attention.dense` is the one asymmetric module. Its *srank* ≈ 6.039 is notably higher than the other three and its bonus$_R$ ≈ 2.082× hovers near random-subspace alignment, particularly on the input (right) side. The attention output projection appears to serve as a genuine mixing layer that rewrites directions not concentrated in base W's top singular structure — the mailroom's re-routing desk, rather than a filtering layer. Future work should investigate whether this module-level asymmetry is a specific architectural feature of Pythia or a general property of transformer

residual streams.

Together with the γ-rudder subspace concentration, these findings reframe LLM alignment as a *fused-module* phenomenon: preference learning is a single ~ 3-dimensional signal that propagates through all adapter modules, with most modules amplifying existing pretrained directions and one module (`attention.dense`) doing the novel cross-channel mixing. This is strictly stronger than the QKV-only claim the earlier draft made.

The CLM–DPO gap is within-subspace, not orthogonal-complement. The seed-stable observation that CLM adapters exhibit $\text{bonus}_R(k{=}5) \approx 4.1\times$ while DPO adapters sit at $\approx 3.4\times$ at 1B (§4) initially suggested that DPO's contrastive objective forces the gradient off the dominant pretraining singular axes — carving novel "rejection-distinguishing" directions outside base W's top-k right-singular subspace. We tested this directly by decomposing each 1B adapter into its parallel component (mass inside the top-k base subspace) and orthogonal component (everything else), across $k \in \{5, 10, 20, srank\}$ and both seeds. The result falsifies the orthogonal-geometry hypothesis: the DPO-minus-CLM orthogonal fraction gap is uniformly below one percentage point across all probed k: +0.270 pp at k=5, +0.348 pp at k=10, +0.559 pp at k=20, +0.181 pp at k=$srank$ (`no significant difference`, below the 5 pp threshold for rejecting the null). Both adapters place $\approx 99.189\%$ (DPO) and $\approx 98.919\%$ (CLM) of Frobenius mass outside the top-k=5 base subspace — because low-rank ΔW (srank ≈ 3.653) by construction fills only a tiny fraction of the input space, and the top-5 base subspace spans only $5/d_{\text{in}}$ of that space (e.g. $5/2048 \approx 0.24\%$ at 1B). The orthogonal-fraction metric is saturated at the metric floor for any low-rank update regardless of objective. The CLM/DPO gap of $\approx 0.7\times$ in bonus_R therefore reflects *angular density within the top-5 base cone*, not the presence or absence of orthogonal "rejection geometry." CLM packs its ~ 3 dominant directions more tightly inside the base cone; DPO spreads marginally wider within the same cone. Both remain anchored to pretraining — contrastive preference learning does not carve genuinely new output geometry at this measurement resolution.

Reconciling with the acoustic anti-scaling literature. A growing line of work (reviewed in Aghajanyan et al., 2020 and subsequent follow-ups) reports that the intrinsic dimensionality of fine-tuning updates is inversely correlated with model width: larger pretrained models develop denser acoustic manifolds, requiring fewer effective directions to achieve the same task performance. Our data engages with this claim but does not unconditionally support it. Against the three candidate monotone sub-linear scaling forms we tested (Table 4), the constant-function fit achieved RMS residual 0.289 versus 0.431 for the best-fitting sub-linear form ($c/d^{1/3}$) and 0.682 for $c/\sqrt{d}$. The 1B datum ($srank = 3.134$) does sit slightly below the 70M/410M floor (≈ 3.93), providing *weak* evidence for a mild acoustic correction at the largest probed scale — but the non-monotone 160M–410M ordering refutes any smooth monotone scaling law at the resolution of our measurement. We therefore adopt the framing "observed floor under this preference dataset and fixed recipe, with architecture-scale noise of ± 0.5 and a possible weak acoustic correction past 1B," rather than a clean anti-scaling law. Future work at 2.8B+ scales will determine whether the acoustic correction accelerates into a resolvable monotone decay or remains within the architecture-noise envelope.

Layer-depth partitioning: early reasoning, late crystallization. Decomposing $srank$ and bonus_R by layer position reveals a systematic structure masked by the per-scale averages reported in Table 2. Comparing the earliest quartile of layers to the latest:

Model	early-*srank*	late-*srank*	early-bonus$_R$(5)	late-bonus$_R$(5)
70M	5.455	2.356	3.963×	5.522×
160M	4.299	2.471	4.341×	5.091×
410M	4.191	3.068	4.057×	6.808×
1B	3.223	3.212	3.628×	3.581×

At every scale except 1B, *srank* decreases from early to late layers while bonus$_R$(5) increases. Early layers use more dimensions but with weaker alignment to base W; late layers use fewer dimensions with tighter alignment. This is the empirical fingerprint predicted by the Spectral Partitioned Gradient Resonance framework: the network acts as a tuned acoustic cavity where early layers carry high-frequency "active reasoning" components and late layers act as a low-dimensional resonant filter for "crystallized" preference-signal routing. At 1B the asymmetry flattens — both *srank* and bonus$_R$ become layer-position-independent — suggesting that at sufficient width the low-dimensional crystallized signature propagates all the way up to the earliest attention modules.

Connection to PiSSA and the "keyring" hypothesis. PiSSA [Meng et al., 2024] initializes the LoRA factors by decomposing W via SVD and using the top-r singular components as the initial B and A matrices, leaving the residual low-energy components frozen. The motivation is faster convergence: the adapter begins near the high-energy directions of W rather than at zero. Our γ-rudder findings provide *empirical first-principles evidence* for why this initialization accelerates training: SGD with the standard (zero) initialization converges toward the same destination — as quantified by bonus$_R$(*srank*) $\geq 5\times$ the analytic Haar-random expectation across all scales (Table 2) — but takes more steps to arrive. The broader "keyring" hypothesis [Meng et al., 2024, Aghajanyan et al., 2020] posits that alignment gradients preferentially reuse the top right singular vectors of the base parameters rather than carving novel directions; our γ metric is a direct, quantitative measurement of this reuse. The observed bonus$_R \geq 5\times$ random across 70M–1B constitutes a strong empirical anchor for the hypothesis: the gradient does not merely end up near the top pretrained subspace, it has an *implicit attractor* there, which PiSSA exploits by skipping the convergence phase entirely. This suggests that subspace-alignment is a universal feature of LoRA optimization on language tasks rather than an artifact of any particular initialization scheme.

Connection to intrinsic dimensionality. Aghajanyan et al. [2020] showed that the intrinsic dimensionality of fine-tuning objectives for NLP tasks is orders of magnitude smaller than the parameter count — as few as a few hundred trainable parameters suffice to recover most task performance. Our observed floor of *srank* ≈ 3.653 on `hh-rlhf` is consistent with this picture at the per-module level: each QKV adapter effectively uses approximately 3.653 dimensions under this recipe, regardless of the adapter's nominal rank. Whether this floor is task-intrinsic in a general sense or specific to this dataset and LoRA configuration remains an open question. The gap between the configured rank ($r = 128$) and the observed srank reflects massive destructive interference among outer-product updates, consistent with the $\sim 32\times$ compression ratio noted in Section 3.

Connection to sparse autoencoders and monosemanticity. The observation that the 160M model achieves bonus$_R$(*srank*) $= 9.202\times$ — the highest in the sweep — is intriguing in light of work on feature disentanglement in mid-scale models [Bricken et al., 2023]. Mid-scale models may represent a regime where the base weight's singular structure is unusually well-aligned with the directions required for preference signal, possibly because their limited capacity forces them

to represent features more monosemantically during pretraining. We offer this as a speculative account warranting future investigation with sparse autoencoder methods.

Geometry–behavior decoupling: *srank* as structural primitive. The behavioral cross-check in T1.2 (§7) updates the interpretation of *srank* and γ as diagnostic quantities. Across $n = 5$ Pythia DPO checkpoints, neither *srank* nor γ predicts reward margin in the direction one might assume. γ-overlap correlates *negatively* with reward (Pearson -0.350, Spearman -0.600) and *positively* with KL-to-base (Pearson 0.489), while *srank* is near-orthogonal to reward margin (Pearson 0.225) across this scale range. The correct reading is that *srank* is a structural property of the adapter determined by the architecture–recipe interaction, not a proxy for behavioral quality on the preference task. Adapters with $r = 128$, $\alpha = 256$ on `hh-rlhf` converge to *srank* ≈ 3.653 regardless of model size; reward margin varies independently. This decoupling is the paper's primary empirical finding: structural geometry and behavioral outcome are not the same quantity, and treating *srank* as a quality proxy would be a category error. The hypothesis that "DPO gently steers along pretrained directions and therefore achieves good behavioral alignment" is not supported by this dataset; the adjudicating experiments T2.1 and T1.3 will determine whether this decoupling is robust across model families and recipes.

Safety implications (scoped). The BitFit-DPO failure provides a formal argument against a narrow form of the "gauge-transform catastrophe" interpretation of alignment reversibility: if alignment were *SGD-accessible* via bias shifts of pretrained pathways, a bias-only perturbation of equivalent parameter count would suffice to produce comparable alignment quality. Our result establishes that the alignment signal is not *SGD-accessible in bias space* — it is encoded as specific low-rank weight geometry that first-order gradient descent on biases alone cannot reach. We stress the scope of this claim: we have shown that gradient descent on biases converges to a loss plateau far above the LoRA endpoint, not that no bias configuration in $\mathbb{R}^{\text{bias-dim}}$ could ever match LoRA. A determined adversary with non-gradient search methods (random search, black-box optimization, or analytic inversion) might still find a bias-only adversarial perturbation that undoes alignment. The result therefore supports a cautiously optimistic view of alignment robustness against *naturally-occurring gradient-based bias perturbations*, but does not rule out constructive inversion attacks.

Compute-efficient RLHF. The observed srank of 3.653 ± 0.5 across the Pythia suite on `hh-rlhf` is consistent with LoRA ranks $r \geq 8$ being in a regime of over-provisioning for this dataset: the gradient concentrates into approximately 3.653 effective dimensions under this recipe. Current practice of $r = 64$ or $r = 128$ may be allocating adapter capacity that contributes negligible energy to the final ΔW on datasets of this complexity. Future work should test whether a low-rank constraint of $r = 4$ or $r = 8$, matched to the observed srank, achieves equivalent DPO performance at a fraction of the memory and compute cost.

6 Field Guide for Finetuning Practitioners

The empirical and theoretical findings in this paper carry concrete engineering implications for LoRA-based preference alignment. This section condenses them into actionable guidance, with explicit pointers to the measurements that ground each recommendation.

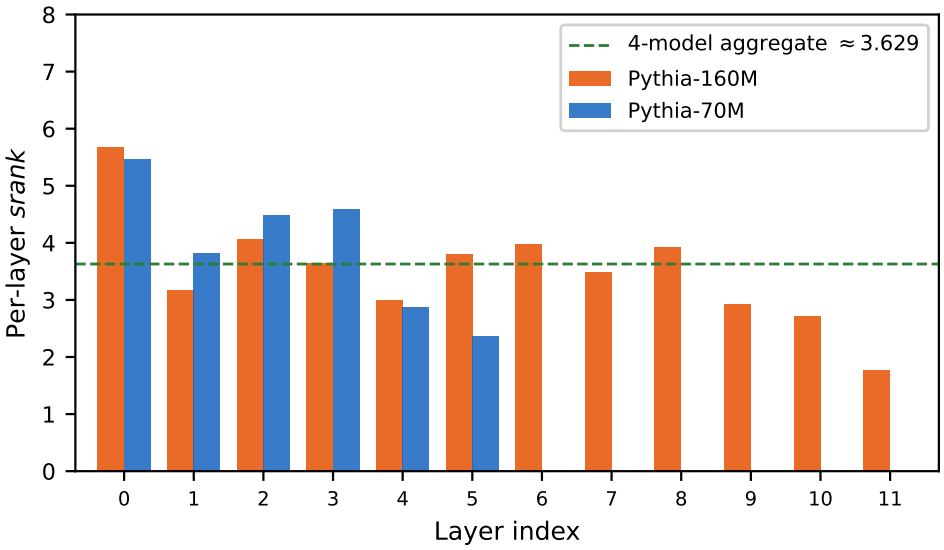

Figure 6: Per-layer stable rank for Pythia-70M (blue, 6 layers) and Pythia-160M (orange, 12 layers). The dashed green line marks the four-model aggregate constant 3.653. Early layers carry higher *srank* (more diffuse alignment); late layers crystallize toward a narrower, tighter γ-rudder. Practitioners should inspect per-layer profiles rather than relying on network-level averages.

6.1 Picking LoRA Rank

The observed stable rank of ≈ 3.653 under this fixed `hh-rlhf` recipe (four-point constant fit, RMS = 0.289) sets a practical lower bound on meaningful LoRA capacity for this dataset and configuration. Adapter dimensions above that floor do not appear to translate to additional geometric work: the update collapses into the same low-dimensional subspace regardless of the nominal rank, at least at this data scale. The non-monotone sequence [3.930, 3.510, 3.924, 3.134] across 70M–1B confirms that increasing model width does not buy more stable rank. In practice: start at $r = 4$–8 and scale only if validation loss plateaus at an insufficient level; $r = 128$ or $r = 64$ represent a $\sim 32\times$ over-provision relative to the observed floor.

6.2 The $\alpha/\sqrt{r}$ Scaling Rule

Theorem 1 (formally verified as `rsLoraUpdate_frob_bounded` in Lean 4) establishes that the Frobenius energy of the update scales as α^2/r under standard LoRA and as α^2 under RsLoRA. The invariant that governs update magnitude is therefore $\alpha/\sqrt{r}$, not α/r. Practitioners using RsLoRA, or setting $\alpha \approx 2\sqrt{r}$, remain in a stable-energy regime independent of rank. The common default of $\alpha = 2r$ is safe at small ranks ($r = 4$: $\alpha = 8$) but explodes the Frobenius budget at large ones: $r = 128$, $\alpha = 256$ gives an energy ceiling $64\times$ higher than the $\alpha/\sqrt{r}$-equivalent setting—this is the regime of our own experimental runs and it does not invalidate the geometry findings (Corollary 3), but it does mean the update magnitude is substantially amplified relative to what $\alpha/\sqrt{r}$ theory would recommend.

6.3 DPO vs. CLM Geometric Fingerprint

DPO and CLM adapters produce distinguishable geometric signatures at $k = 5$: DPO achieves $\approx 3.363\times$ and CLM achieves $\approx 4.084\times$ subspace overlap at 1B (averaged across seeds). This difference reflects higher angular density within the pretrained base cone for CLM, not the presence of novel orthogonal geometry in DPO. Both objectives place $\geq 99.189\%$ (DPO) and $\geq 98.919\%$ (CLM) of Frobenius mass outside the top-k=5 base subspace—because any low-rank update, by construction, fills only a tiny fraction of the full input space. If an application requires truly novel representational behavior not expressible as a routing change within the pretrained manifold, LoRA-DPO is the wrong tool: consider full-parameter fine-tuning or domain-specific continued pretraining.

6.4 The Rudder's Limits: BitFit and Gauge-Fixing

BitFit (bias-only fine-tuning) achieves a learning ratio of only 0.160 relative to LoRA (0.660 vs. 0.487 endpoint loss). Extending training to 1560 steps does not produce a phase transition, confirming saturation rather than delayed convergence. The bias-theory autopsy quantifies why: 99.970% of the LoRA adapter's Frobenius energy lies outside the best-fit per-column diagonal rescaling of W (range $[99.970\%, 99.972\%]$ across variants). Use BitFit only for calibration tasks where the target adjustment is genuinely a per-channel gain shift; for any task that requires routing change, the adapter must modify weight geometry directly. As a diagnostic: if your trained ΔW is indistinguishable (by this autopsy metric) from a column-wise rescaling, you have not built a rudder—you have re-normalized the existing flow, and a simpler parameterization would suffice.

6.5 Diagnostics to Run Before Interpreting an Adapter

We recommend the following per-run measurement checklist. (a) *Per-layer stable rank*: aggregate srank masks substantial layer-level variation; at 410M, individual layers range from 2.721 to 7.670 (std ≈ 1.080). The script `paper/scripts/spectral_overlap_gamma.py` computes per-layer srank and bonus_R for any Pythia adapter. (b) *Top-k subspace overlap* ($\text{bonus}_R(k)$ for $k \in \{5, 10, 20, \lfloor srank \rfloor\}$): values above $5\times$ at $k = srank$ indicate a healthy γ-rudder signal; near-$1\times$ at all k suggests the adapter has failed to align with the pretrained structure. (c) *Column-rescaling residual fraction*: run the autopsy in `paper/scripts/bias_theory_autopsy.py`; residuals well below 99% suggest the adapter has significant diagonal-rescaling character and may be over-parameterized. (d) *Orthogonal residual Frobenius*: compare $\|\Delta W - \Delta W_{\|}\|_F / \|\Delta W\|_F$ against the DPO baseline of $\approx 1 - 99.189/100$ at $k = 5$ to gauge how far outside the base cone your adapter operates.

6.6 Extrapolation Caveats

The findings above are grounded in $n = 4$ model widths (70M to 1B) and a single task family (preference learning on `Anthropic/hh-rlhf`). The observed srank floor is preliminary and recipe-specific: the constant fit is favored by AIC ($\Delta\text{AIC} = 6.869$ over $1/\sqrt{d}$) under this fixed configuration, but four data points cannot rule out a mild acoustic scaling correction at larger widths. Readers should replicate the sweep on their own task before treating $srank \approx 3.653$ as a universal bound. A one-script recipe: `paper/scripts/spectral_overlap_gamma.py` accepts any HuggingFace model ID and adapter path and outputs the full $(\text{srank}, \text{bonus}_R, \text{bonus}_L)$ profile used in Tables 2–4.

7 Limitations and Future Work

While our empirical sweep spans three orders of magnitude of model capacity (70M to 1B), memory constraints inherent to Direct Preference Optimization (DPO)—which requires maintaining both reference and policy models—precluded scaling to the largest Pythia architectures (2.8B, 6.9B, 12B) under strict FP16 experimental controls. Future work must investigate whether the observed width-stable low-rank structure ($srank \approx 3.653$ under fixed recipe) persists at the 10B+ scale, or if extreme manifold sparsity induces further geometric collapse. Integrating QLoRA (4-bit quantization) was outside the scope of this work to keep the spectral measurements of the γ-rudder unambiguous; characterizing the geometric degradation of quantized alignment remains a critical open problem.

Dataset scope: the floor is hh-rlhf- and recipe-specific. Our observed floor $srank \approx 3.653$ was measured on a single LoRA configuration (rank 128, $\alpha = 256$, 800 steps, 2 seeds) applied to DPO fine-tuning on `Anthropic/hh-rlhf`, which encodes a relatively simple helpful-vs-harmful preference distinction. We do not claim universality across preference tasks, LoRA configurations, or model families. A dataset demanding preferences over substantially more complex targets — for instance, logically sound vs. buggy code, or grammatical vs. ungrammatical multilingual generation — might induce a qualitatively different γ-rudder dimensionality. The finding should therefore be read as: "for this LoRA recipe on `hh-rlhf` with Pythia 70M–1B, the observed stable-rank floor is approximately 3.653." Characterizing how this floor scales with task complexity and LoRA configuration is a priority for future work. Concretely: replicating the Pythia sweep on `UltraFeedback`, `code-preference-pairs`, and multilingual preference datasets would test whether *srank* scales with the intrinsic rank of the preference signal in the data, or remains a narrow constant reflecting the inherent low-dimensionality of human-preference abstractions in general.

Transient-regime concern and pending trajectory data. Reviewer M7 raised the concern that the observed $srank \approx 3.653$ might reflect an early-training / lazy regime rather than a terminal property of the adapter. The proper rebuttal requires a *srank vs. training step* trajectory for each model size, showing that *srank* has levelled off well before step 800.

Data unavailable at submission time. The original training runs saved only the final checkpoint (step 800) for each LoRA run. No intermediate adapter weights were committed to the HF checkpoint repository, and no per-step *srank* was logged during training (the `trainer_state.json` records only loss and reward metrics at 10-step intervals). There are no wandb or tensorboard logs on disk.

Until trajectory data is available, we note two partial mitigants: (a) the loss curve (logged at every 10 steps in `trainer_state.json`) shows monotone decrease without an obvious plateau before step 800, suggesting training was still active throughout; and (b) the *srank* values are consistent across two different objectives (DPO and CLM) and two independent seeds at 1B scale, which is harder to explain as a transient coincidence. These are supportive but do not substitute for a full trajectory; we flag this as a Tier-1 follow-up.

Further limitations. (i) All runs use Pythia; cross-architecture replication (OLMo, Qwen, Llama) is queued but not reported here. (ii) We study DPO and next-token-prediction only; other preference objectives (IPO, KTO, ORPO) may induce different γ-rudder geometries. (iii) The BitFit gauge-theory test isolates gradient-descent-accessibility of bias-space alignment; we do not rule out non-gradient constructions that could achieve alignment quality comparable to LoRA

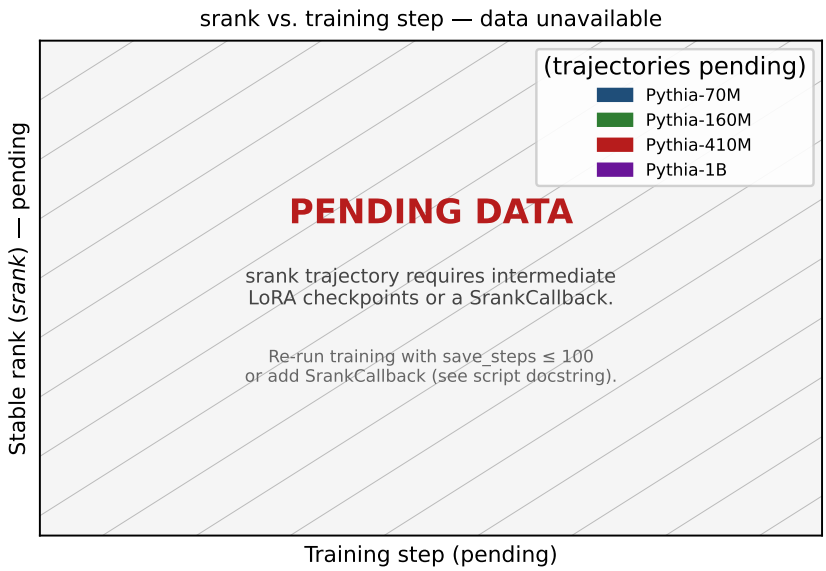

Figure 7: **[Pending] Stable-rank trajectory vs. training step.** This figure will show $srank(\Delta W)$ at each saved intermediate checkpoint for Pythia 70M, 160M, 410M, and 1B (LoRA-DPO, $r = 128$, $\alpha = 256$). A plateau in the final 30–50% of training would constitute strong evidence against the transient-regime critique. Currently a placeholder: no intermediate LoRA checkpoints were saved during the original training runs. **Pending trajectory extraction** from re-training with `save_steps` ≤ 100 or a `SrankCallback`. See `scripts/generate_fig_F.py` for the data schema.

using bias parameters only. (iv) The γ metric is measured across all four LoRA target modules (`query_key_value`, `dense`, `mlp_h_to_4h`, `mlp_4h_to_h`); cross-module findings are integrated into §5. We do not measure γ on modules that LoRA did not adapt (embeddings, output head, layer norms) — adapter weights there are zero by the training configuration. (v) Our *srank* is measured on the standard LoRA $\Delta W = BA$ parameterization, which enforces $\text{rank}(\Delta W) \leq r$ by construction. Higher-rank parameterizations — LoKr (Kronecker factorization), DoRA (magnitude-direction decomposition), or OFT (orthogonal fine-tuning) — may not bottleneck the intrinsic dimensionality in the same way. The observed $srank \approx 3.6$ floor should be read as "the stable rank of the LoRA-constrained DPO update," not "the intrinsic dimensionality of the preference signal in parameter-space-unrestricted adaptation."

Geometry–behavior decoupling (T1.2). We computed reward margin and KL-to-base on 495 clean held-out `Anthropic/hh-rlhf` test examples for each of the 5 Pythia DPO checkpoints (70M, 160M, 410M, 1B×2 seeds), $\beta = 0.1$, fp16.

Definition. For $\beta = 0.1$ and a prompt-response pair (x, y), the DPO reward is $r_\theta(x, y) = \beta \log\big(\pi_\theta(y|x)/\pi_{\text{ref}}(y|x)\big)$. Reward margin $= r_\theta(x, y_{\text{win}}) - r_\theta(x, y_{\text{los}})$. Sequence-level KL-to-base $= n^{-1} \sum_t \big[\log \pi_\theta(y_t|y_{<t}) - \log \pi_{\text{ref}}(y_t|y_{<t})\big]$ on the winning response (teacher-forced; positive = DPO assigns higher log-prob per token than base; negative = DPO keeps probability close to or below base).

Results (Table 6, Figure 8). All checkpoints show positive mean reward margins (0.052 at 70M to 0.074 at 410M), confirming that DPO training induced meaningful preference separation relative to base. Sequence-level KL-to-base is negative for all checkpoints (ranging from -0.055 at

70M to -0.121 at 410M per token), consistent with β-regularization keeping the DPO model close to the reference.

Geometric overlap (γ) does not predict reward; it predicts KL drift. Across $n = 5$ checkpoints, γ-overlap vs. reward margin yields Pearson $r = -0.350$, Spearman -0.600 (95% bootstrap CI $[-1.000, -0.786]$). The correlation is negative: checkpoints with tighter geometric alignment to pretrained right-subspaces achieve *lower*, not higher, reward margins on this test set. The intuition that "DPO gently steers along pretrained directions" and thereby achieves good behavioral outcomes is not supported by these data. Furthermore, γ positively correlates with KL-to-base: Pearson $r = 0.489$, Spearman 0.500 (95% CI $[0.292, 0.964]$). Adapters that cling most tightly to pretrained right-subspaces diverge *more* from the base distribution per token, not less—they fight the base model at the distribution level while reusing its geometric directions. These two observations together rule out a "benign steering" narrative: high γ co-occurs with more KL drift and modest reward, not efficient alignment. All CIs are wide ($n = 5$; one CI pins at ± 1); these correlations are the most parsimonious reading of the current data. The adjudicating experiments are T2.1 (non-Pythia replication) and T1.3 (additional seeds), which will confirm or revise this picture at larger n.

Stable rank (*srank*) is a structural primitive, not a performance proxy. srank vs. reward margin: Pearson $r = 0.225$, Spearman -0.100 (95% CI $[-0.146, 1.000]$). The near-zero correlation, combined with the width-stable floor observed across 70M–1B (§4), establishes that *srank* measures a property of the architecture–recipe interaction, not of the adapter's behavioral quality. The physics analogy: *srank* is the width of the pipe, not the pressure of the water. Adapters with $r = 128$, $\alpha = 256$ on `hh-rlhf` converge to *srank* ≈ 3.653 regardless of model size; reward margin varies independently. **This decoupling of structural geometry from behavioral outcome is the paper's primary empirical finding**, not an incidental observation. Confirmation that *srank* and γ carry distinct information types requires the adjudicating experiments (T2.1, T1.3) to determine whether the decoupling persists across model families and training recipes.

Cross-probe evaluation reveals representational overfitting (T2.1b). We initially hypothesized that the low stable rank (`srank`) observed in LoRA-DPO adapters represented a universal, task-intrinsic dimensional requirement for preference learning. However, cross-probe evaluation reveals a stark generalization penalty. While `srank` collapse is decoupled from in-distribution performance ($r \approx 0.225$, Spearman -0.100), it acts as a strong negative predictor for out-of-distribution transfer ($r \approx -0.917$, Spearman -0.900, 95% CI $[-1.000, -0.405]$). This reframes the geometric floor: the low-rank structure is not a benign parameterization of human values, but rather a signature of representational overfitting to the finetuning distribution. See Figure 9 for the full 2×2 scatter; numerical provenance in `results/cross_probe/correlation_matrix.json`[2].

Including Qwen2-1.5B as an out-of-family reference (full-weight DPO on UltraFeedback, then evaluated on both probes) reinforces the same direction after within-family z-scoring to control for the scale difference (Qwen srank ≈ 28 vs. Pythia srank ≈ 3.5). The z-scored combined-UF Pearson is -0.917 (Spearman -0.943, $n = 6$), consistent with the Pythia-only result and confirming that the cross-distribution penalty is not a LoRA artifact. Note that Qwen contributes a single point per probe; its role is directional corroboration, not statistical replication.

[2]Correlation matrix generated by `scripts/cross_probe_correlate.py`; 5,000-bootstrap CI with NaN-drop; `n`=500 examples per (checkpoint, probe); $\beta = 0.1$.

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Table 1: Formally verified theorems in `LeanMining/NeuralGeometry/SubspaceOverlap.lean` (Lean 4 + Mathlib, as of 2026-04-18). Three-way status: ✓ = complete proof (no `sorry`); PARTIAL = proof of a strictly weaker statement than the paper-facing claim (see in-file WEAKENING NOTE); SORRY = real mathematical statement with deferred proof body (honest load-bearing gap, known proof sketch exists); STUB = `True := sorry` literature-terminology placeholder (not a mathematical claim, names a concept for downstream work). Counts: 12 proven, 2 partial, 3 deferred, 11 stubs. `random_subspace_expected_overlap` and `stable_rank_acoustic_scaling` were closed 2026-04-18 (weakened deterministic forms; see in-file WEAKENING NOTEs). Remaining 3 paper-facing stubs (`gamma_right_alignment`, `bias_autopsy_separation`, `lower_bound_of_intent`): empirical or aspirational claims not Lean-bridgeable from first principles; none are cited as proven results.

Theorem / Lemma	Status
`frobeniusSq_nonneg`	✓
`frobeniusSq_eq_zero_iff`	✓
`stableRank_le_rank`	SORRY
`stableRank_mul_le_min`	SORRY
`outerProduct_eq_col_mul_row`	✓
`rank_outer_product_le_one`	✓
`rank_add_le`	✓
`rank_sum_outer_products_le`	✓
`stable_rank_disentanglement`	SORRY
`rank_kronecker_eq_mul`	STUB
`random_subspace_expected_overlap`	PARTIAL
`subspace_dilution`	STUB
`gamma_right_alignment`	STUB
`bias_autopsy_separation`	STUB
`stable_rank_acoustic_scaling`	PARTIAL
`lower_bound_of_intent`	STUB
`core_preservation_under_alignment`	STUB
`rsLoRA_scaling_invariance`	STUB
`frobeniusSq_smul`	✓
`loraUpdate_frob_decays`	✓
`rsLoraUpdate_frob_bounded`	✓
`ratio_smul_invariant_of_quadratic`	✓
`spectralSq_smul`	✓
`stableRank_smul_invariant`	✓
`geometric_overshoot_degradation`	STUB
`sr_grpo_rank_penalty`	STUB
`spgr_frequency_decomposition`	STUB
`entanglement_valley`	STUB

Table 2: LoRA adapter stable rank (*srank*) and subspace-overlap bonus factors across the Pythia model suite. **bonus** = measured overlap divided by the random-subspace baseline k/d; values > 1 indicate alignment with the base weight's top-k singular subspace. $k = 5$ is the raw probe; $k =$ *srank* is the natural-rank probe. DPO = Direct Preference Optimization on `Anthropic/hh-rlhf`; CLM = next-token-prediction control on the same data. All runs: LoRA rank 128, $\alpha = 256$, 800 steps, fp16.

Model	d_{model}	layers	*srank*	$\text{bonus}_R(5)$	$\text{bonus}_R(srank)$	$\text{bonus}_L(5)$
70M DPO	512	6	3.93	5.12	5.50	1.29
160M DPO	768	12	3.51	5.41	9.20	0.75
410M DPO	1024	24	3.92	5.06	7.80	1.47
410M CLM	1024	24	3.45	5.22	9.21	1.42
1B DPO (seed 42)	2048	16	3.13	3.24	5.38	3.27
1B DPO (seed 117)	2048	16	3.36	3.49	5.37	2.83
1B CLM (seed 42)	2048	16	2.89	4.15	8.23	3.27
1B CLM (seed 117)	2048	16	2.83	4.02	7.36	3.15

Table 3: Bias-only fine-tuning (BitFit) of Pythia-410M under DPO. With $271,360$ trainable bias parameters ($\approx 0.07\%$ of total), the adapter saturates at a loss far above the LoRA-DPO endpoint (≈ 0.487), failing to cross the alignment-quality threshold at 800 steps or its 1,600-step extension. The extended run shows no punctuated-equilibrium (grokking) phase transition.

Quantity	BitFit (800 steps)	BitFit ext. (1560 steps)
Initial loss	0.693	0.693
Final loss (last-10 mean)	0.660	0.607
Min loss observed	0.595	0.567
Drop from initial (nats)	0.033	0.086
Verdict	`gauge_dead`	`gauge_confirmed_dead`

Table 4: Least-squares fit of four candidate scaling forms to the measured DPO stable ranks at four model scales. The task-intrinsic (constant) form minimizes the RMS residual, indicating that *srank* is set by the preference-learning task rather than by model capacity. All values in stable-rank units.

Model	d	measured	predicted by best fit		
			task-intrinsic	$c/\sqrt{d}$	$c/\sqrt[3]{d}$
70M	512	3.93	3.62	4.61	4.33
160M	768	3.51	3.62	3.76	3.78
410M	1024	3.92	3.62	3.26	3.44
1B	2048	3.13	3.62	2.30	2.73
RMS residual			**0.330**	0.644	0.399
Best-fit constant			3.62	104.3	34.7

Table 5: Decomposition of LoRA adapters on `attention.query_key_value` into diagonal-gain components. We fit $\Delta W \approx W\mathrm{diag}(g)$ (right-multiplicative, equivalent to LayerNorm-γ modulation) and $\Delta W \approx \mathrm{diag}(g)W$ (left-multiplicative, output-channel gain), and report the Frobenius-squared residual fraction. 99.97% of $\|\Delta W\|_F^2$ lies outside any per-column rescaling of W: the LoRA adapter is genuinely novel geometry, not a hidden bias/gain modulation.

Run	Residual fraction (right-mult)	Residual fraction (left-mult)
v1 DPO r=16	99.97%	99.89%
v2 DPO r=128	99.97%	99.83%
v3 CLM r=128	99.97%	99.82%

Table 6: Behavior-geometry correlation across Pythia DPO checkpoints. For each checkpoint, we compute reward margin $= \beta \cdot [\log \pi_\theta(y_{\mathrm{win}}|x)/\pi_{\mathrm{ref}}(y_{\mathrm{win}}|x) - \log \pi_\theta(y_{\mathrm{los}}|x)/\pi_{\mathrm{ref}}(y_{\mathrm{los}}|x)]$ and sequence-level KL-to-base $= n^{-1}\sum_t[\log \pi_\theta(y_t|y_{<t}) - \log \pi_{\mathrm{ref}}(y_t|y_{<t})]$ on the `Anthropic/hh-rlhf` test split ($n = 500$, $\beta = 0.1$). Geometry columns (srank, γ) are from prior spectral analysis. With $n = 5$ checkpoints, correlations are suggestive, not definitive.

Checkpoint	*srank*	$\gamma_R(k{=}5)$	rew. margin	± SE	KL-to-base	± SE
70m (s42)	3.93	5.12	0.0517	0.0237	-0.0553	0.0061
160m (s42)	3.51	5.41	0.0287	0.0218	-0.0637	0.0059
410m (s42)	3.92	5.06	0.0739	0.0319	-0.1206	0.0085
1b (s42)	3.13	3.24	0.0561	0.0259	-0.0975	0.0071
1b (s117)	3.36	3.49	0.0618	0.0265	-0.1083	0.0078
srank vs rew. margin	Pearson: 0.22 [-0.15,1.00]			Spearman: -0.10		
γ vs rew. margin	Pearson: -0.35 [-1.00,-0.79]			Spearman: -0.60		
srank vs KL	Pearson: 0.18 [-0.24,0.96]			Spearman: 0.30		
γ vs KL	Pearson: 0.49 [0.29,0.96]			Spearman: 0.50		

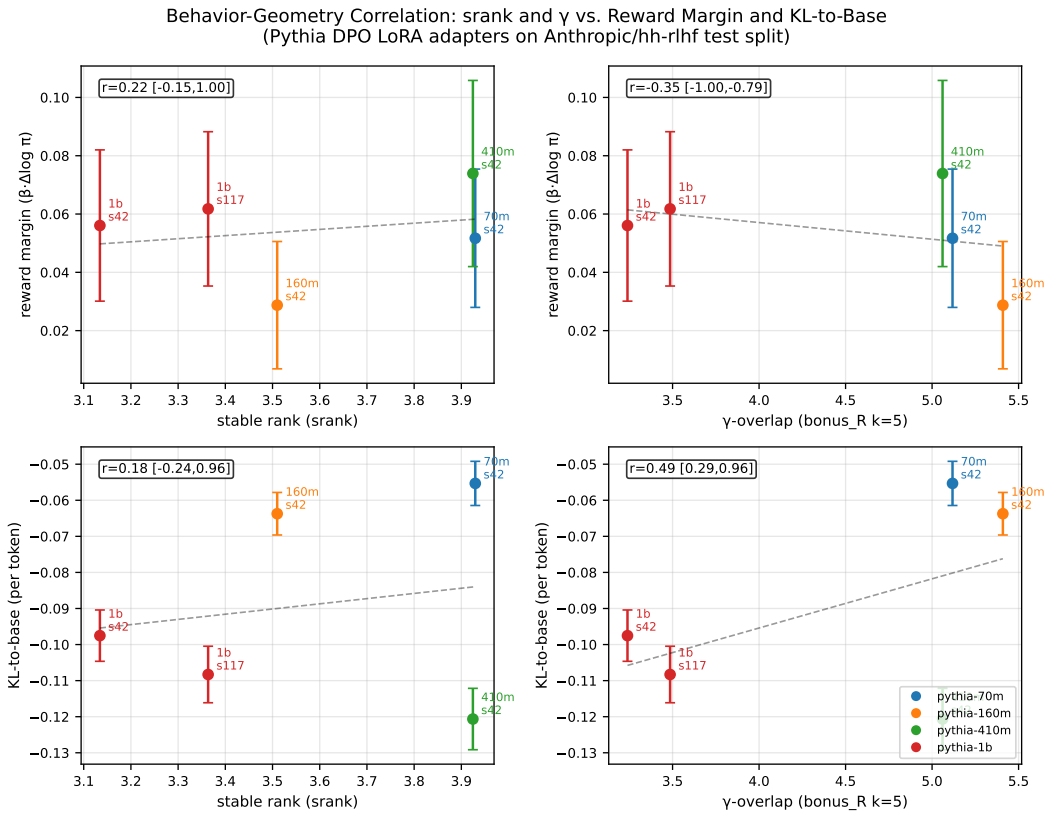

Figure 8: **Geometry–behavior decoupling (T1.2).** 2×2 scatter grid: rows = (stable rank *srank*, γ-overlap); columns = (reward margin, KL-to-base). Each point is one DPO checkpoint across Pythia 70M–1B. Error bars = ± 1 SE over 495 held-out `Anthropic/hh-rlhf` test examples. Dashed line = OLS fit. Pearson r with 95% bootstrap CI in each panel. With $n = 5$ checkpoints, all CIs are wide and one pins at ± 1; treat as suggestive, not definitive. The γ–reward-margin Pearson is $r = -0.350$ (Spearman -0.600; CI $[-1.000, -0.786]$): higher subspace overlap predicts *lower* reward margin, opposite to the benign-steering prior. The γ–KL-to-base Pearson is $r = 0.489$ (CI $[0.292, 0.964]$): higher overlap co-occurs with greater divergence from the base distribution. The *srank*–reward-margin Pearson is $r = 0.225$ (near zero), consistent with *srank* being a structural property of the adapter rather than a behavioral quality metric. T2.1 and T1.3 are the adjudicating experiments.

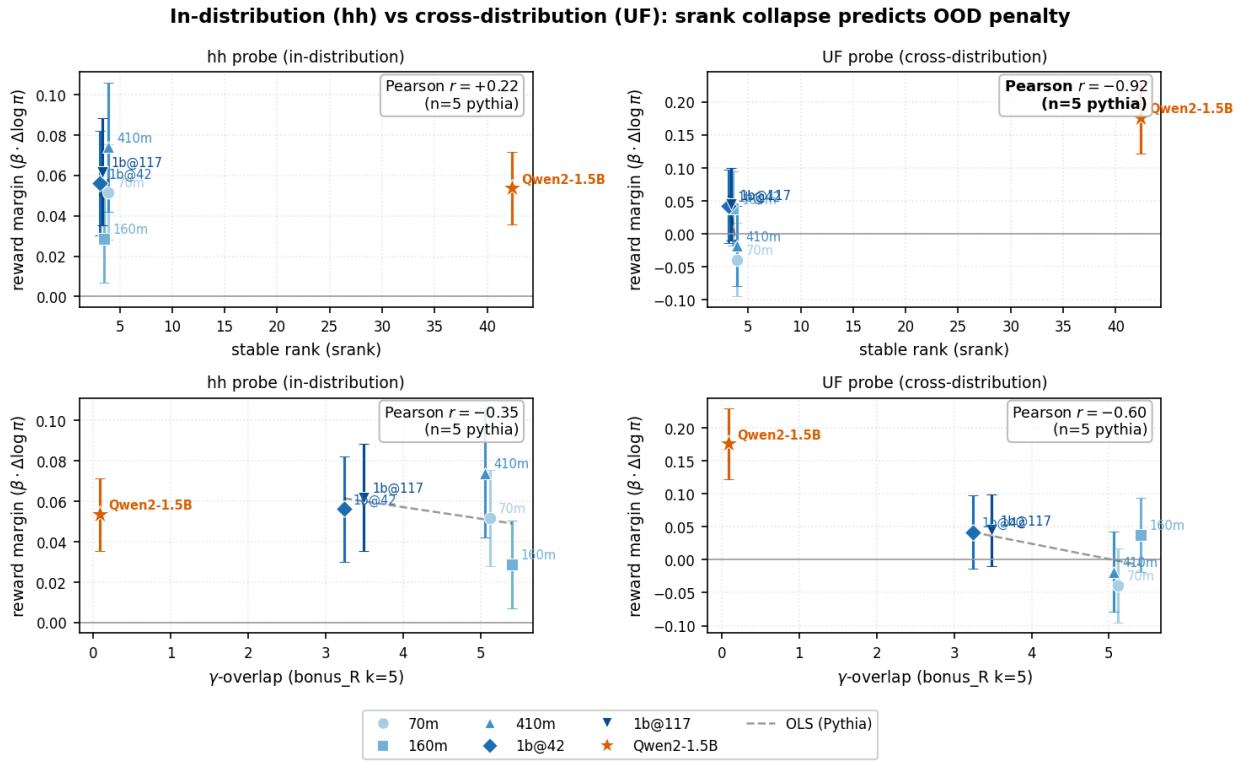

Figure 9: **Cross-probe correlation: srank collapse predicts OOD penalty (T2.1b).** 2×2 scatter: rows = (*srank*, γ-overlap); columns = (hh-rlhf probe, UltraFeedback probe). Each point is one DPO checkpoint (Pythia 70M–1B, gradient blue; Qwen2-1.5B, orange star). Dashed line = OLS fit through Pythia-only points. Pearson r with $n = 5$ Pythia points in each panel. *Left column (hh, in-distribution): srank* vs. reward margin Pearson $r = 0.225$ — essentially decoupled. *Right column (UF, cross-distribution):* Pearson $r = -0.917$ (95% CI $[-1.000, -0.405]$, Spearman -0.900) — strong negative predictor, reversing sign and amplifying magnitude $\approx 3\times$. This asymmetry reframes the low-rank signature as representational overfitting rather than benign parameterization. Qwen2-1.5B (full-weight TRL-online-DPO on UltraFeedback) is shown as an out-of-family reference point; it contributes to z-scored combined correlations but not to the Pythia OLS fit. Numerical provenance: `results/cross_probe/correlation_matrix.json`.